University Course Registration & Prerequisite Validation Engine

«actor»
Student

«actor»
Admin

View weekly schedule

Search course catalog

View course details
and prerequisites

Register for a section

Join waitlist

Drop a section

Sign in

Manage prerequisite
relationships

View section roster
and waitlist

Manage courses
and sections

«extend»
(section full)

«include»

«include»

«include»

«include»

«include»

«include»

«include»

«include»

Detect schedule clash

Allocate seat
(concurrency-safe)

Verify credit limit

Validate prerequisites

Promote first eligible
waiting student

Detect prerequisite cycle